

Integrating a COE file into the BRAM of an AMD FPGA device in Vivado 2025

Checking the maximum size of COE files

Design Sources (2)

- Exbinder02(Exbinder02_Full) (ExBinder06.vhd) (4)
 - myNetWorkLevel00 : NetWorkLevel00(NetWorkLevel00_Full) (NetWorkLevel00_00.vhd) (4)
 - u1 : Neuron_BramInLine(Neuron_BramInLine_all) (Neuron_BramInLine_pkg.vhd) (4)
 - myNetWorkLevel01 : NetWorkLevel01(NetWorkLevel01_Full) (NetWorkLevel01_00.vhd) (4)
 - myNetWorkLevel02 : NetWorkLevel02(NetWorkLevel02_Full) (NetWorkLevel02_00.vhd) (4)
 - myNetWorkLevel03 : NetWorkLevel03(NetWorkLevel03_Full) (NetWorkLevel03_00.vhd) (4)
- Coefficient Files (1)
- Constraints
- Simulation Sources (2)
- Utility Sources

Source File Properties

Neuron_BramInLine_pkg.vhd

Enabled

Location: F:/PH_01/ULg_Cours/AC 2017-20XX-Phd/Code VHDL AC/VHDL/10-MNI

```
107 -- Variables environnement - Gestion global
108 signal valdata : std_logic_vector (0 to 789);
109 signal convadd : std_logic_vector(16 DOWNTO 0);
110
111 -- Variable pour le processus de traitement de la tâche
112
113 -----
114 -- Déclaration des Fonctions system -----
115
116
117 begin
118   BramROM_ip_instance : BramROM_ip
119     port map (
120       clka => CLK,
121       addra => convadd,
122       douta => valdata
123     );
124
125 -----
126 -- Processus Lecture BRAM -----
127
128 CycleLint: process (CLK, Reset, go)
129   --variable local pour le processus de traitement de la tâche
130
```

Upper address

Block memory name

IP Catalog

ExbinderMNIST01 - [F:/PH_01/ULg_Cours/AC 2017-20XX-Phd]

File Edit Flow Tools Reports Window Lay

Flow Navigator

- PROJECT MANAGER
 - Settings
 - Add Sources
 - Language Templates
 - IP Catalog
- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design
- SIMULATION
 - Run Simulation
- RTL ANALYSIS
 - Run Linter

PROJECT MANAGER

Sources

- Design Sour
 - Exbin
 - myl
 - myl
 - myl
 - myl
 - Coeffici
 - Constraints
 - Simulation S

Hierarchy IP S

Source File Propri

BramROM_ip.xc

Double-click to start the "IP Catalog" setup.

Bram on Block Memory Generator

The screenshot shows the Vivado IDE interface. The **IP Catalog** tab is active, displaying a search for "bram" with 3 matches. The results table is as follows:

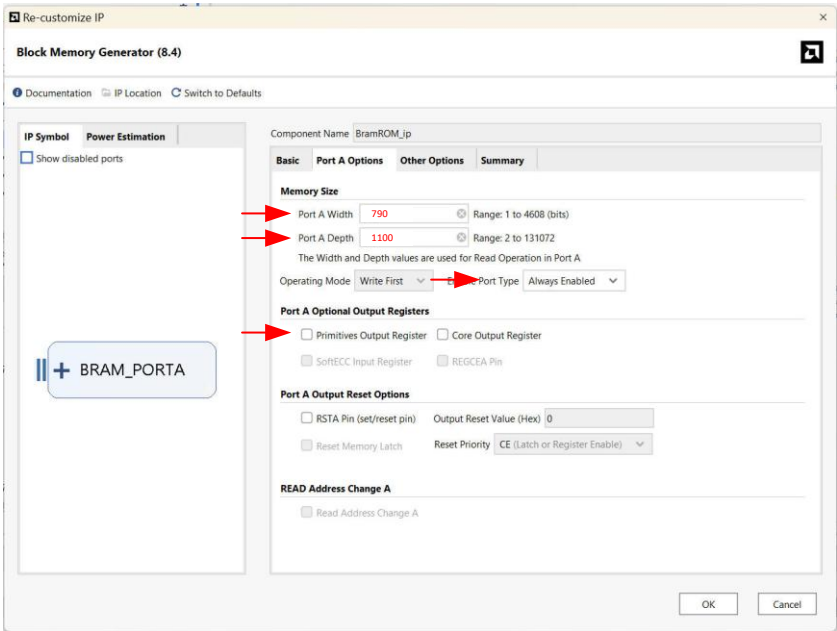
Name	Category	Status	License	VLN
Vivado Repository				
Embedded Processing				
Memory and Memory Controller				
AXI BRAM Controller	AXI4	Production	Included	xilinx.com:ip:axi_bram_ctrl:4.1
LMB BRAM Controller	AXI4	Production	Included	xilinx.com:ip:lmb_bram_ctrl:4.0
Memory Storage Elements				
RAMs & ROMs & BRAM				
Block Memory Generator	AXI4	Production	Included	xilinx.com:ip:blk_mem_gen:8.4

The **Design Runs** tab is also visible, showing the status of the synthesis and implementation runs. The synthesis run is completed, and the implementation run is in progress.

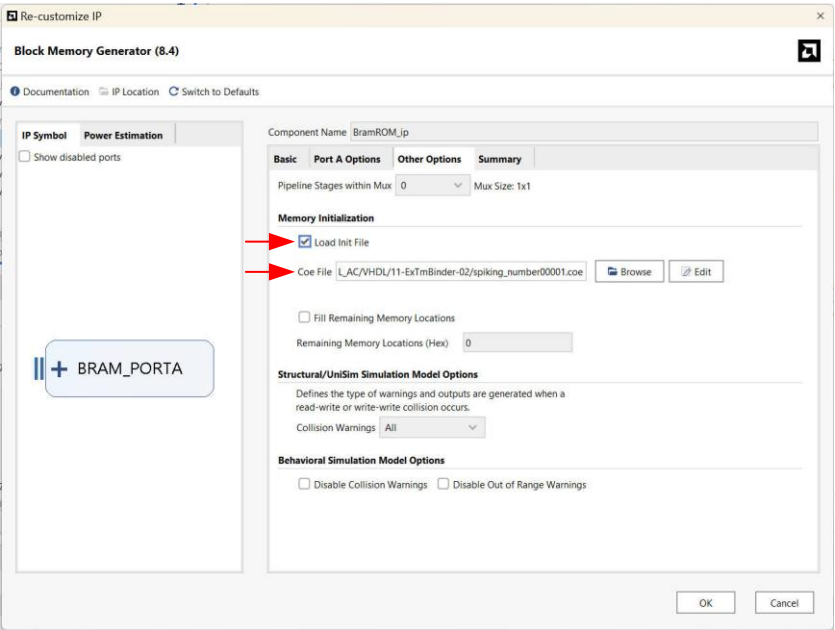
Step 1 – Component Name and Memory Type

The screenshot shows the **Block Memory Generator (8.4)** configuration window. The **Component Name** is set to **BramROM_ip**. The **Memory Type** is set to **Single Port ROM**. The **Interface Type** is set to **Native**. The **ECC Options** are set to **No ECC**. The **Write Enable** options are set to **Byte Write Enable** and **Byte Size (bits)** is set to **9**. The **Algorithm Options** are set to **Minimum Area** and **Primitive** is set to **8x2**.

Step 2 – Size of Component



Step 3 – Load COE File



Step 4 – Final

